JavaScript Lab Manual – Lab 2

Course: Web Engineering

Software Required:

- Visual Studio Code
- Node.js
- Google Chrome

Task 1 – Arrays in JavaScript

Objective

Understand how to create and manipulate **arrays** in JavaScript.

Theory

An **array** is used to store multiple values in a single variable.

Arrays are useful when working with collections of data such as student marks, product lists, or user information.

Example

```
let fruits = ["Apple", "Mango", "Banana"];
console.log(fruits);
console.log(fruits[0]);
fruits.push("Orange");
console.log(fruits);
```

Output

```
Apple
Mango
Banana
Orange
```

Example – Loop Through Array

```
let numbers = [10,20,30,40];
for(let i = 0; i < numbers.length; i++){
console.log(numbers[i]);
}
```

Tasks

1. Create an array containing **5 subject marks**.
2. Print all marks using a **for loop**.
3. Calculate the **sum of all marks**.
4. Calculate the **average marks**.
5. Add a new mark using the push() method.

Task 2 – Objects in JavaScript

Objective

Understand how to store structured data using **objects**.

Theory

An **object** stores data in **key-value pairs**.

Example structure:
key : value
Objects are commonly used to represent **real-world entities**, such as students, cars, or products.
Example

```
let student = {
name: "Ali",
age: 21,
department: "Software Engineering"
};
console.log(student.name);
console.log(student.age);
console.log(student.department);
```

Adding New Property

```
student.cgpa = 3.5;
console.log(student);
```

Tasks

1. Create an object for **student information** with the following properties:
    - Name
    - Age
    - Department
    - CGPA
2. Print all object properties using console.log().
3. Add a new property **University** to the object.
4. Display the complete object.

Task 3 – DOM Basics
Objective
Learn how JavaScript interacts with **HTML elements using the DOM (Document Object Model)**.
Theory
DOM allows JavaScript to:

- Access HTML elements
- Modify text content
- Change styles
- Create or remove elements

Example – Changing Text
HTML

```
<!DOCTYPE html>
<html>
<body>
```

```
<h2 id="title">Welcome Students</h2>
<button onclick="changeText()">Click Me</button>
<script src="script.js"></script>
</body>
</html>
```

JavaScript

```
function changeText(){
document.getElementById("title").innerHTML = "Welcome to JavaScript DOM Lab";
}
```

Example – Changing Style

```
document.getElementById("title").style.color = "blue";
```

Tasks

1. Create a webpage with a **heading and a button**.
2. When the button is clicked:
    - Change the **heading text**.
3. Change the **heading color** using JavaScript.

Task 4 – JavaScript Events

Objective

Understand how JavaScript handles **user interactions**.

Theory

An **event** occurs when a user interacts with a webpage.

Examples:

- Clicking a button
- Typing in an input field
- Submitting a form
- Moving the mouse

JavaScript can respond to these events.

Example – Button Click Event

HTML

```
<input type="text" id="name" placeholder="Enter your name">
<button id="btn">Submit</button>
<p id="output"></p>
```

JavaScript

```
document.getElementById("btn").addEventListener("click", function(){
let name = document.getElementById("name").value;
document.getElementById("output").innerHTML = "Hello " + name;
});
```

Tasks

Create a simple **interactive webpage** with the following features:

1. Add an **input field** where the user enters their name.
2. Add a **Submit button**.
3. When the button is clicked:
    - Display the message

Hello <User Name>, Welcome to Web Engineering Lab

4. Change the text color of the message using JavaScript.

Bonus Challenge (For Advanced Students)

Create a **Student Marks Calculator** that:

1. Takes **5 subject marks as input**.
2. Stores them in an **array**.
3. Calculates:
    - Total marks
    - Average marks
4. Displays **Pass or Fail using an if condition**.

Expected Learning Outcomes

After completing this lab students will be able to:

- Work with **arrays and objects**
- Use **loops with arrays**
- Understand **DOM manipulation**
- Handle **user events in JavaScript**
- Build **interactive webpages**